A PORTFOLIO FOR THE SENSES

Amy Hu.

Making things people can touch, see, and taste, and testing ideas in the lab

touch → see → taste → hear → think



01 / TACTILE BOOKS



03 / BAKING & PHYSICS



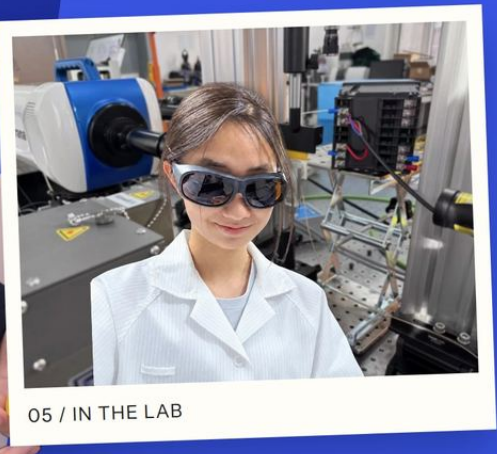
02 / MINIATURE HOUSE



MADE FOR THE SENSES.



04 / ON STAGE



05 / IN THE LAB

CONTENTS

SIX WAYS

Each project began with a sense, and ended with something I could hand to someone else.

AMY HU · 2026

01 TOUCH

Even Without Sight

Tactile picture books in felt and braille for blind children, the visits and the exhibition.



P. 03 – 08

02 SIGHT

A House in Miniature

A furnished home small enough to sit on a desk.



P. 09

03 TASTE

The Physics of Baking

A book I wrote: twelve months, twelve recipes and the science behind each one.



P. 10 – 11

04 SOUND

Hearing the Past & Echoes Recovered

A voice-built memory space for older people, and 103 century-old recordings restored.

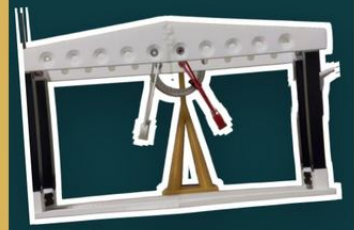


P. 12 – 13

05 MIND

Research & Entrepreneurship

Physics tournaments, an AI hearing aid, a startup, metallic glass, optics and modelling.



P. 14 – 22

06 MUSIC & MOTION

Guzheng & Taekwondo

Performing on stage and competing on the mat.



P. 23

TOUCH

Even without sight, there is still vision.

Care & Reading, our school reading-tutor club of more than 100 members, works with ABF, a Beijing NGO for visually impaired children. Every week, each volunteer tutors one blind child online, and the same pairs stay together year after year. I led 30+ members to design and hand-sew tactile picture books in felt and braille, ran 10+ visits, made an exhibition and a short film, and raised over CNY 2,000.

MY ROLE Activities leader & organiser. I designed the books, planned every visit and sewed pages with the team.

50+ long-term reading pairs, every week

30+ members sewing tactile books

10+ visits to the children

¥2k+ raised for materials



every page is read with the fingertips ↓



reading by touch, on one of our visits



SKETCHBOOK · THE KINGDOM OF SHAPES

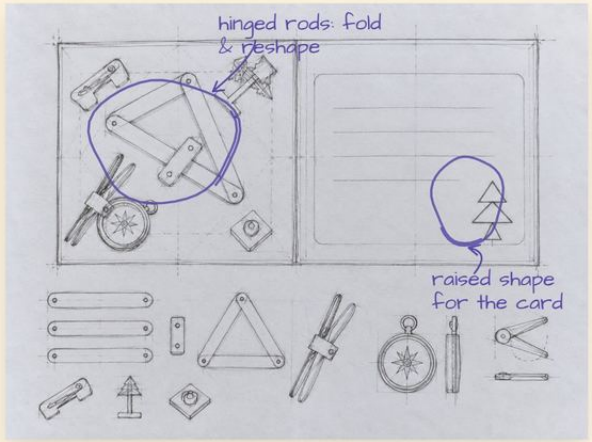
Design drawings for the tactile books

Each book is a story set in the Kingdom of Shapes: Jianjian the triangle crosses town to a New Year concert, and Fangfang the square grows into a family of shapes. Before any felt was cut, I drew every spread in pencil: the layout, the loose parts, the joints in section and a cutting list, so the page could be built exactly and safely for small hands.

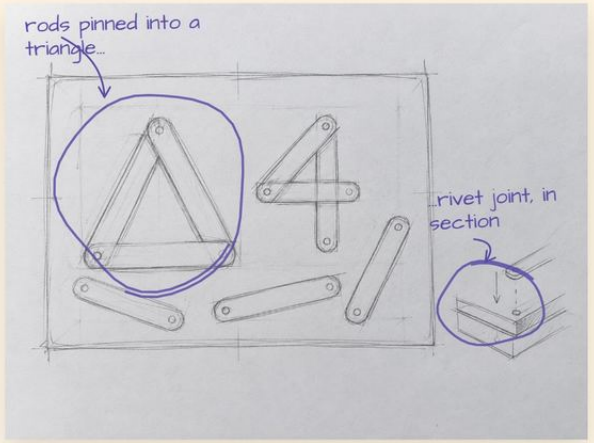
FOUR RULES I DESIGNED BY

RAISED	every shape in felt strips, metal or stitched edges
PAIRED	every scene faces a braille card
ALIVE	something spins, rings or lights up
SAFE	no loose ends, every small part sewn

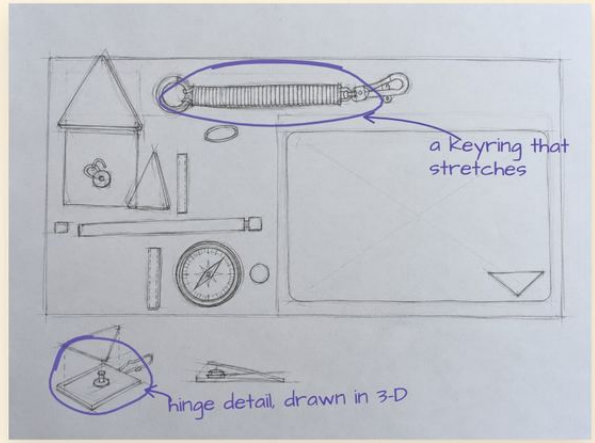
SKETCH 01 — SPREAD PLAN



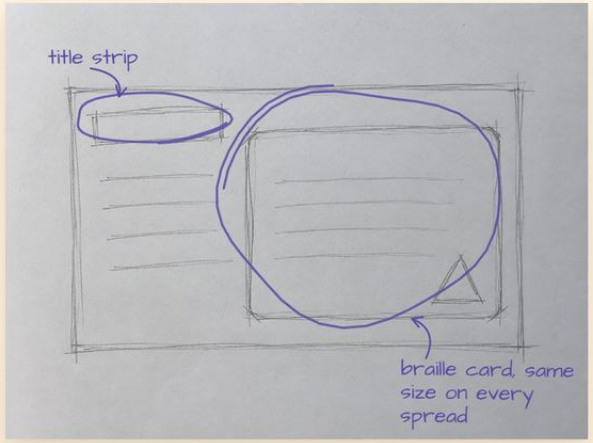
SKETCH 02 — RODS & JOINT



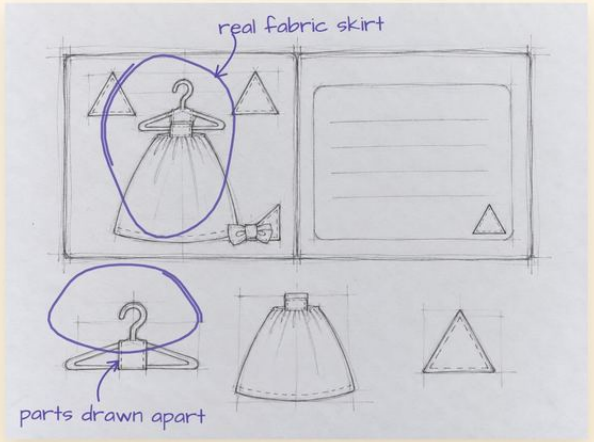
SKETCH 03 — ROOF, KEYRING, COMPASS



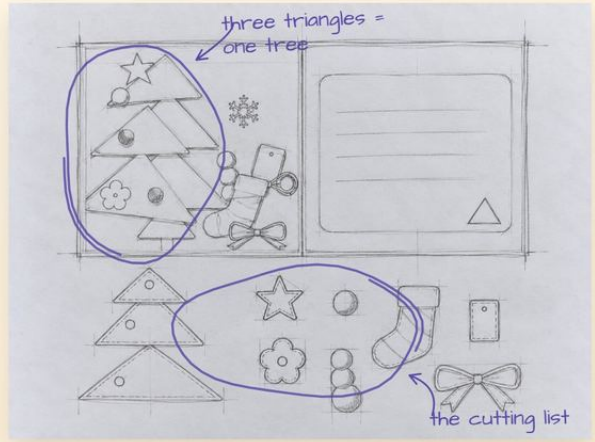
SKETCH 04 — PAGE GRID



SKETCH 05 — THE DRESS



SKETCH 06 — THE TREE



HOW A PAGE IS MADE



00 · The kit

Everything for one page: felt sheets, coloured felt strips, a braille story card and thread. We packed a kit like this for every club member.



00 · The bench

Where the books were cut and sewn, and where I build robots and science projects: pegboard tools, a 3D printer and a laptop for the design files.

01



Measure & mark

Shapes are ruled onto the felt before anything is cut.

02



Cut the felt

Each scene piece is cut by hand.

03



Sew the strips

Felt strips are stitched down to make sides a finger can trace.

04



Lay out the scene

Pieces are placed and checked before sewing.

05



Fix the braille card

The story card is fixed opposite the scene.

06



The finished book

Every scene raised in felt, with its braille card opposite, ready for a child's fingers.

The BOOKS

Two of our three books, both set in the Kingdom of Shapes. Each spread pairs a felt scene with a braille story card, and the raised-line icon on the card tells the child which shape to find next. Book 1 hides a triangle in every scene; in Book 2 one square turns into something new on every page.

BOOK 1 · JIANJIAN THE TRIANGLE

8 SPREADS · TRIANGLES OF EVERY KIND



01 · Triangle Town

"My name is Jianjian. I live in Triangle Town."

Craft sticks form loose triangles a child can lift and rearrange; the red roof is a raised felt triangle.



02 · The horn

"Toot toot! No. 3 Obtuse Street, a letter!"

A real toy horn, a compass and a coiled cord give the street its sounds and textures.



03 · The invitation

"My gown! My heels!" A cone-shaped gown.

A felt envelope with stitched edges opens onto a wooden cone; a metal clef hints at the concert.



04 · The gown

"Honk honk!" Cars in a right triangle.

A floral fabric skirt hangs on a real hanger between felt triangles, so the gown can be lifted and felt.



05 · Traffic jam

A Christmas tree glows at the hall door.

Craft-stick roads, toy cars and road signs, all sewn down so nothing comes loose.



06 · The tree

An equilateral conductor keeps equal beats.

Stacked felt triangles make the tree, decorated with buttons, a sock and a snowflake.



07 · The triangle

A waiter carries fizzy drinks.

A real metal triangle and beater: the child can strike it and hear the shape.



08 · Countdown

"10, 9, 8… 1!" The New Year bell rings.

A cup, a ladder and stick fireworks on blue felt; the bell icon matches the New Year bell.

BOOK 2 · FANGFANG'S TRANSFORMATIONS

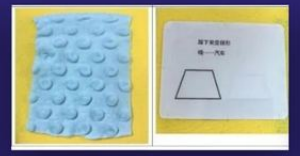
8 SPREADS · ONE SQUARE, EIGHT NEW SHAPES



01 · Fangfang

Meet Fangfang, a little square who loves to change shape.

A yarn-haired face with button eyes opens the book and introduces Fangfang.



02 · Flying carpet

She flies up into a parallelogram: whoosh, a flying carpet!

Bubbled fabric gives the carpet a bumpy surface to explore under the fingertips.



03 · Car

She crouches into a trapezoid: vroom, a car!

A felt racetrack, a traffic light and a toy fire engine that can be rolled along.



04 · Rattle drum

She gets cross and puffs up round: boom, a drum!

A wooden pellet drum wrapped in felt rattles when the handle is twisted.



05 · Tree

Into the forest, layer on layer: a big tree.

Square felt leaves with real buttons on a twig branch can be buttoned and unbuttoned.



06 · Flower

She spins into a hexagon: a flower!

A felt flower with a button centre stands in a glittered pot.



07 · School bag

Pieced together, neat and square: a school bag.

Real backpack buckles on a felt bag let children practise clipping them together.



08 · Gem

A somersault, and she is a rhombus: a gem!

A stretchy pop tube coiled into a ring can be pulled, bent and squeezed.

Felt scene
Raised shapes to trace with a finger.

Braille card
The story line, in braille and print.

Raised-line icon
Tells the child which shape comes next.

Something alive
A top that spins, a drum that booms, lights that glow.

THE VISITS

We have visited the children more than ten times. Each visit starts with ice-breaker games, then a workshop such as *My Little Planet*, where children paint with materials they can feel. We play, read the books together and interview the NGO teachers, and we filmed one visit as a short documentary.

WHAT CHANGED FOR THE CHILDREN

50+ children learn beyond what their schools currently teach, one-to-one, every week.

1:1 Consistent, not one-off: each child keeps the same volunteer week after week.

∞ Long-term volunteers who also play with the children, not just teach them.



The volunteer team on a visit to the NGO.



The children read the books with their fingertips.



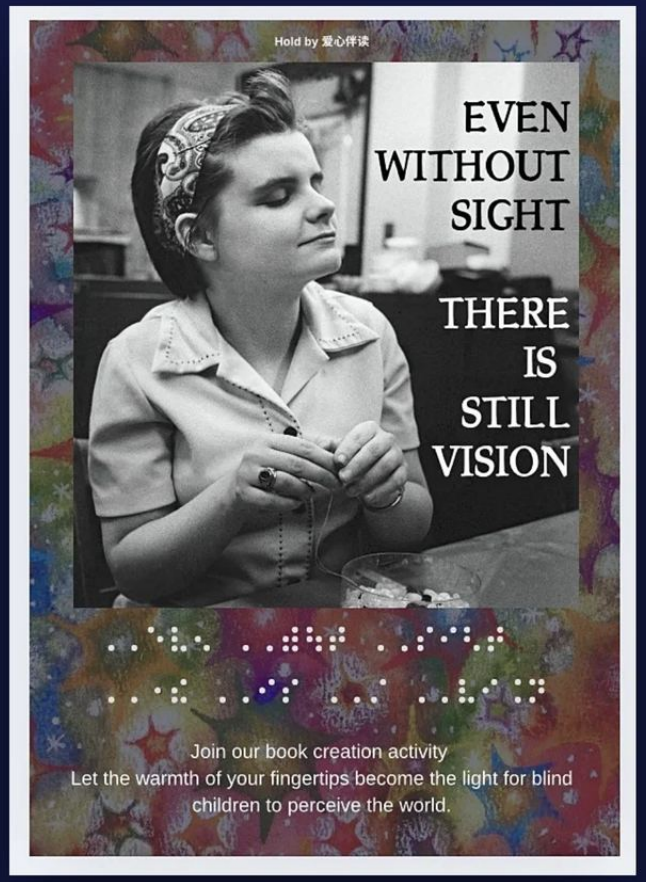
My Little Planet: paintings made with textures instead of colours alone.



TAG, TRANSLATED

"Thank you, Hu, and the volunteer team of No. 101 Middle School. May this 'Fuba' doll of love and good fortune bring you all joy!"

A gift from the NGO teachers.



The exhibition poster. Its line reads: "Let the warmth of your fingertips become the light that helps blind children perceive the world."

EVEN WITHOUT SIGHT, THERE IS STILL VISION

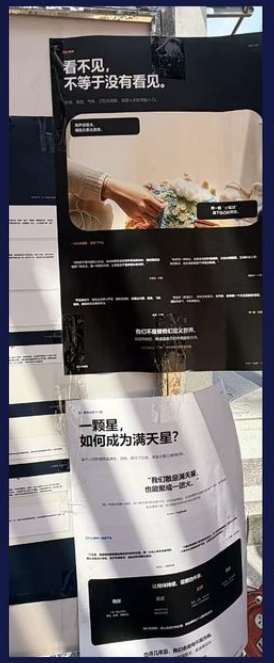
We closed the project with a club exhibition. Classmates picked up the books and learned how blind people move through and perceive the world, a group most of them had never thought about.

INSIDE THE EXHIBITION

Five hands-on stations: **braille basics**; the **shoes and canes** blind people use; **facts about blindness**; a **screening** of our documentary; and a table of **tactile materials**.

60+

VISITORS OVER A WEEK













WHAT VISITORS WROTE

"I like that you let visitors touch and interact instead of simply reading about blindness."

— a teacher

"The tactile books are thoughtful, but you could explain more clearly how each texture helps a blind child understand the story."

— a classmate

"Consider adding feedback from visually impaired users themselves to strengthen the exhibition."

— a classmate

"It made me wonder what everyday objects would feel like if I could only understand them through touch."

— a classmate

SIGHT

A house in miniature

The tactile books asked how a story can be read by hand. This project asks the opposite question: how much can a world hold when it is read only by the eye? I shrank a whole home onto a desktop: five rooms on one floor, every piece made by hand.

01
Plan the floor
Five rooms on one level, with the bed lofted above the kitchen.

02
Build the shell
Walls, floor and loft must be square, or nothing inside sits straight.

03
Make the furniture
Bookcase, desk, bed and sofa, assembled with tweezers and glue.

04
Dress every room
Bedding, a towel, food on the plates: details that make it lived in.

ROOMS	BUILD TIME	HAND-MADE PARTS
5+	10+ hrs	200+



Making it small is not the hard part. The hard part is making every room believable.

The scale has to agree, the surfaces have to look used, and the light has to land the way it would in a real home.

LOOK CLOSER · SIX CORNERS OF THE HOUSE

1



The study
Tiny printed books, a desk lamp and a chair pulled up to work.

2



Dressing room
A yellow towel on the rail, a basin and slippers by the step.

3



Moon window
A round window frames the living room like a garden view.

4



Kitchen
Tucked under the loft bed, every pot in its place.

5



Dining table
Dinner is already served on a patterned runner.

6



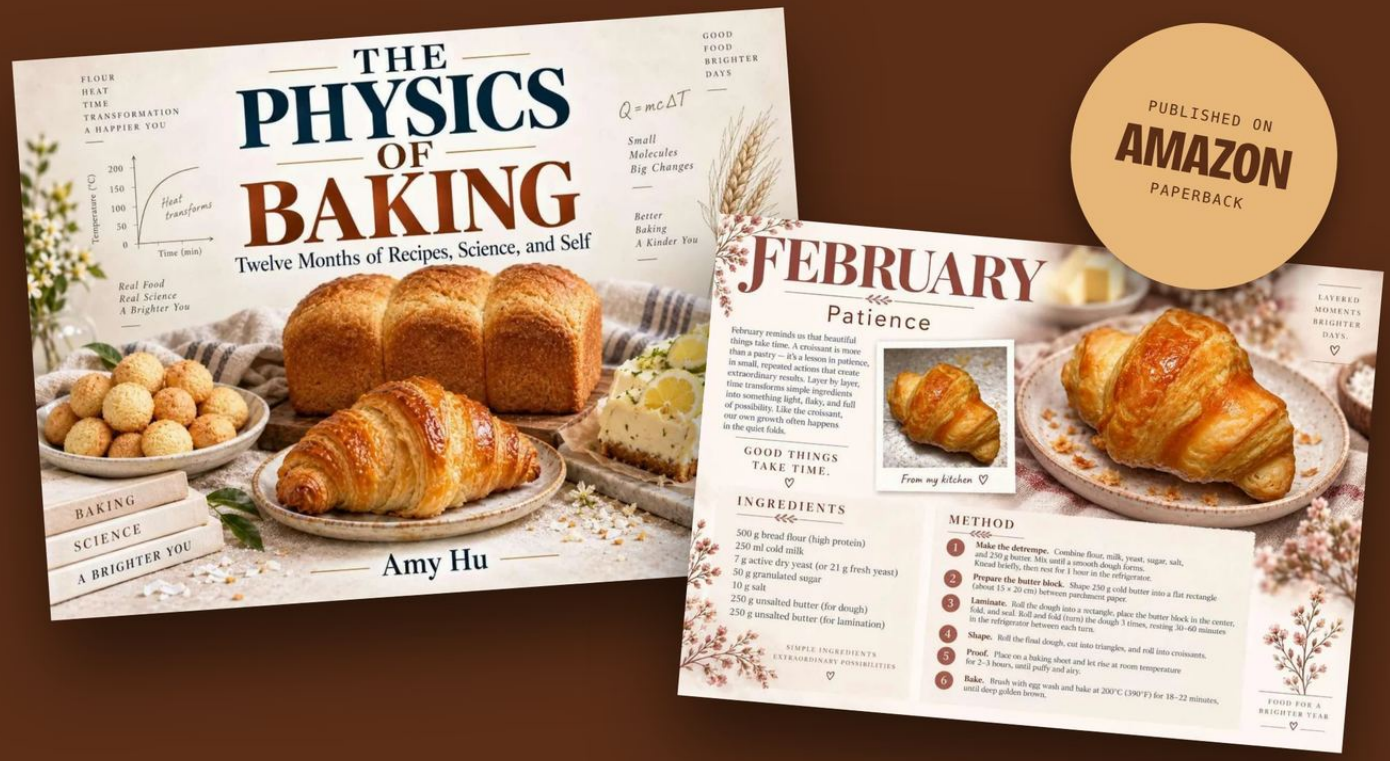
Tea table
A teapot and cups smaller than a fingernail.

THE PHYSICS OF BAKING

Twelve months of recipes, science and self. A book I wrote, and published.

I baked, photographed, wrote and designed all 38 pages myself. Each month pairs one recipe with a page on the physics that makes it work: how whipped egg whites hold air, why folded butter turns into flaky layers, when a crust begins to brown. The finished book is on sale on Amazon.

> *"For my grandfather, whose hearing challenges first taught me to notice the quiet details of everyday life."*
> FROM THE DEDICATION



THE PHYSICS ON EVERY PAGE · WHAT HAPPENS AS THE TEMPERATURE RISES



FROM KITCHEN TO AMAZON

01 Bake & measure **02** Photograph at home **03** Explain the science **04** Design 38 pages **05** Publish on Amazon KDP

ONE YEAR, TWELVE EXPERIMENTS

Every bake in the book is also a small experiment. The inner ring shows each month's bake and the idea its science page explains. The outer ring shows the page that opens that month in the book.

Foams, emulsions, heat transfer, starch and fat: twelve ways physics ends up on a plate.

WHAT CHANGED ACROSS THE YEAR

The year starts simple, with coconut balls in January, and moves through laminated croissants and macarons to a Basque cheesecake that depends on a steep temperature gradient.



INTERACTION · SOUND · AI · DESIGN FOR OLDER ADULTS

HEARING THE PAST

A memory space that older people build by talking.

Reminiscence sessions often start from old photos and objects, but many older people no longer have pictures from those years, and stock images of old Shanghai or old Beijing never match the lane they grew up in. So I reversed the process: instead of showing a ready-made picture, the system builds the place while the person describes it. They only have to talk.

01 **A companion that asks**
An AI companion greets them and follows their story, one question at a time: what did the doorway look like?

02 **The street appears**
As they speak, a 360° street view forms, and the tree, the bicycle and the wonton stall are placed into it.

03 **Sound in its place**
Bicycle bells, street calls and opera from an old radio come from the right direction, fainter with distance and muffled behind walls.

04 **Corrections in plain words**
"No, the wonton seller was at the other end." The stall moves; when unsure, the system asks first.

05 **Stories pinned to places**
When they talk about a person or an event, the recording is attached to that spot in the scene.

06 **Easy to stop**
"Too loud" silences everything within half a second; if they seem tired, the companion offers a rest.





360° SCENES GENERATED BY THE SYSTEM, SHOWN FLATTENED

EXPORTS ARE LABELLED: GENERATED BY AI FROM THE ACCOUNT, NOT REAL PHOTOGRAPHS

Design choices

Voice only. No buttons and nothing to learn; the staff panel sits in a corner the participant does not see.

Rules for the companion. It never invents or corrects a memory, asks one question at a time, and speaks slowly and briefly, so the session never feels like a test.

Period texture. Photographic images rather than illustration, and sound shaped like an old radio, loudspeaker or gramophone.

Ethics & research

Participants are identified by number only, consent comes first, and data stays on the local machine and can be exported or deleted at any time.

Three built-in conditions test whether a co-built scene draws out more memories: **sound only**, **a generic regional scene**, and **a scene built from the person's own account**. Each session produces a report of speaking turns, stories told and scene edits, comparable across sessions.

How it is built

three.js for the 360° view and occluding objects; Web Audio for spatial sound, distance and period filters; Alibaba Cloud speech recognition and synthesis; Qwen to converse and turn speech into scene edits; Wanxiang to generate scenes and objects; depth estimation for occlusion; a Python FastAPI back end that versions every edit.

My role. Concept, interaction flow, the companion's rules, sound and visual style, and the research conditions. I built the full working system with AI coding tools.

LIBRARY OF CONGRESS · NATIONAL JUKEBOX

ECHOES RECOVERED

Hearing the gramophone age again.

An acoustic-era disc sounds muffled because the horn that recorded it lost almost everything above 8 kHz. With my teammate, I repaired and built phonographs, measured exactly what they distort, and designed a way to undo it.

01 Repair & build
Restored an antique Victor soundbox and built a motor-driven turntable to play and record discs.

02 Denoise & EQ
Compared FIRWIN2 filtering with Dynamic Spectrum Matching (DSM); DSM improved sound quality by 34%.

03 Restore & donate
Averaged DSM into one gain curve that repairs discs with no original, restored 103 recordings from 1901–1925 and donated them back.

04 Archive & exhibit
Built Sound Time Map, an interactive archive of every disc, and shared it with 300+ visitors.

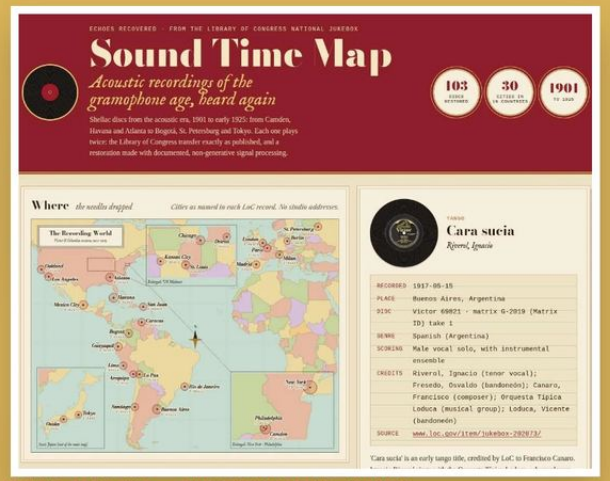

GRAMOPHONE


VICTOR SOUNDBOX, C. 1900S


TEST RIG I BUILT


OPENED & REPAIRED


DIGITISING



SOUND TIME MAP · THE INTERACTIVE ARCHIVE

103
DISCS RESTORED

30
CITIES, 19 COUNTRIES

1901–25
ACOUSTIC ERA

300+
EXHIBITION VISITORS

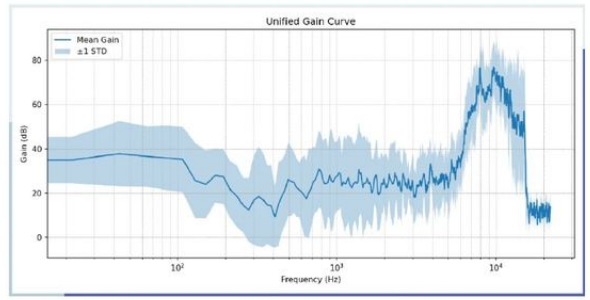


RESULT

34%

better sound quality with DSM than with FIRWIN2.

The averaged curve boosts the highs the horn lost, peaking near 10^4 Hz.



UNIVERSAL GAIN CURVE · MEAN OF DSM REPAIRS

TEAM: AMY HU & SEBASTIAN LIU · RECORDINGS: LIBRARY OF CONGRESS, NATIONAL JUKEBOX

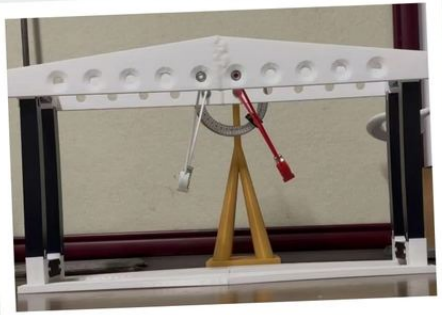
RESEARCH & ENTREPRENEURSHIP

How do people sense the world, and what can engineering do when a sense fails?

Six projects and one startup: from physics tournaments, where I captained my team, through laboratories at Peking University and the Chinese Academy of Sciences, to two assistive products and the company I founded to build them. Each of the next eight pages shows one of them through its paper.



me, on stage



01 P. 15–16

Energy Transport in a Chain of Magnetically Coupled Pendulums

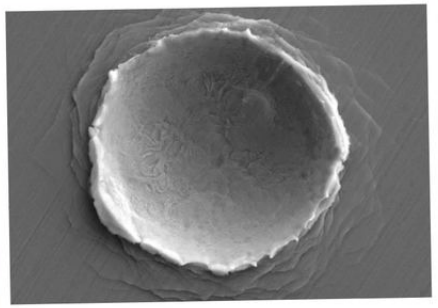
IB EXTENDED ESSAY · IYPT & CYPT TEAM CAPTAIN



02 P. 17

SmartHearing: Scene-Adaptive Hearing Assistance

PEKING UNIVERSITY, INSTITUTE FOR AI · PATENT



03 P. 18

Micro-particle Impacts on Metallic Glasses at High Temperature

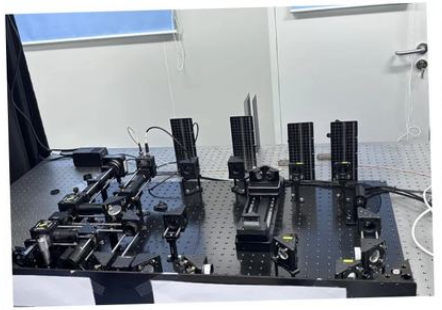
CHINESE ACADEMY OF SCIENCES · FIRST AUTHOR



04 P. 19–20

Curawave Transfer Mattress and the LingTong Startup

CONRAD CHALLENGE · FOUNDER & CTO



05 P. 21

Polarisation Control and Experimental Optics

PEKING UNIVERSITY, INSTITUTE OF PHYSICS



06 P. 22

Mathematical Modelling and Science Publishing

IMMC · HIMCM · 3 FINALIST AWARDS

1
patent for an AI hearing aid

2
first-author research papers

$100k+
raised for my startup

3
modelling Finalist awards

Energy Transport and Localization in a Chain of *Magnetically Coupled Pendulums*

Research question. *How do relevant parameters affect the normal-mode frequencies and thereby influence energy transport and localization in a chain of magnetically coupled pendulums?*

AMY HU · INDEPENDENT RESEARCH · IYPT MACAO TEAM CAPTAIN

ABSTRACT A chain of magnetically coupled pendulums is a macroscopic analogue of systems in which interacting particles exchange and transport vibrational energy: the pendulums are discrete sites, gravity provides an on-site potential, and magnetic repulsion is the nonlinear interaction between neighbours. I measured the normal-mode frequencies of chains of two to five pendulums while varying spacing, magnetic moment, pendulum length and release angle, and compared them with a Lagrangian model built from independently measured parameters. The model reproduced the measured frequencies to within a few percent (mean absolute difference ≈ 1.37%, RMSE 0.045 Hz at $\theta_0 = 30.3°$). The nearly in-phase mode remains controlled by gravity, while higher modes rise with magnetic moment and fall strongly with spacing; larger release angles produce amplitude-dependent softening. Used as a numerical tool, the model shows how the spectrum decides whether energy spreads through the chain, returns to a few modes, or stays localised.

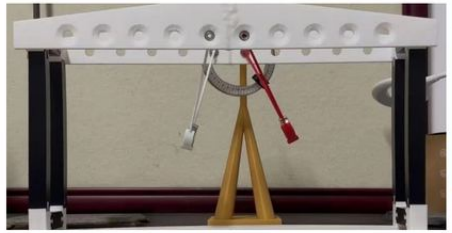


Fig. 1 · My apparatus: PETG brackets set the spacing s; bearings and baskets hold like-oriented magnets.

LEADERSHIP

Captain of the IYPT Macao team. At school I founded our physics lab and community, Wavefront (CYPT / MYPT varsity); I run forums and inter-school exchanges and tutor younger students for tournaments.

1 · MOTIVATION

Unlike microscopic systems, the displacement of every pendulum can be observed directly, while the coupling strength, lattice spacing, damping and initial excitation can be controlled experimentally. The chain lets dispersive wave propagation, modal energy exchange, recurrence and energy localization be studied in an accessible, measurable form.

2 · BACKGROUND

Magnetic repulsion is a contactless, adjustable realisation of a nonlinear oscillator lattice; attached to pendulums, gravity adds an on-site potential and a lower cutoff frequency. Moving breathers have been observed in a ring of magnetic pendulums, and the Fermi–Pasta–Ulam problem showed that energy placed in a few modes may return instead of spreading. The connection between adjustable parameters, the complete eigenfrequency spectrum of a finite chain, and the resulting transport of energy remained insufficiently established.

3 · HYPOTHESIS

Stronger magnetic coupling, from larger magnetisation or smaller spacing, should widen the spectrum and raise most the modes with large relative motion between neighbours. The nearly in-phase mode should stay governed by the gravitational torque.

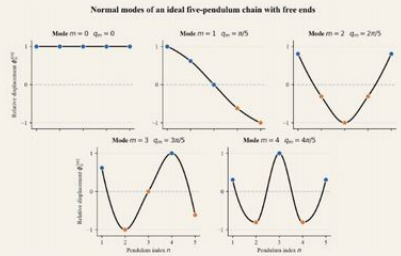


Fig. 2 · Normal modes of a free-end chain, N = 5. As m increases, neighbours differ more and the magnetic contribution raises the frequency.

4 · METHOD

The apparatus used a rigid top rail, sliding PETG brackets, bearing-mounted arms and magnets held in baskets, with a camera normal to the plane of motion. For each setting one end pendulum was released from θ_0, recorded at 30 frames s^{-1} and repeated at least three times; only one variable was changed per series. Each bob was tracked in Tracker, and the eigenfrequencies were extracted by discrete Fourier transform, accepting a peak as a collective mode only when it appeared in more than one pendulum with compatible phases.

The model is a Lagrangian with Rayleigh dissipation: $I_{eq}\,\ddot{\theta}_n + \partial V/\partial\theta_n + \gamma\,\dot{\theta}_n = 0$, with gravity and the measured magnetic pair potential in V.

5 · RESULTS

At $\theta_0 = 30.3°$ the measured frequencies were 1.432, 2.546 and 3.952 Hz, against model values of 1.425, 2.492 and 3.896 Hz: a mean absolute difference of about 1.37% and an RMSE of 0.045 Hz. For N = 3, 4 and 5 the discrepancies were 2.09%, 2.50% and 6.09%, growing with long-range coupling and magnet heterogeneity.

Higher modes rose with magnetic moment and fell strongly with spacing; longer pendulums lowered all frequencies; larger release angles produced amplitude-dependent softening.

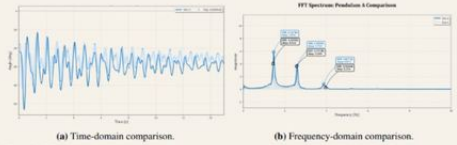


Fig. 3 · Theory vs experiment: simulated peaks near 1.467, 2.600 and 3.867 Hz; measured near 1.435, 2.557 and 3.960 Hz.

6 · CONCLUSION

Spacing, magnetic moment, length, release angle, chain size and damping first set the mode frequencies, bandwidth and nonlinear detuning, which then decide whether energy propagates through several sites, exchanges among a few modes, or becomes spatially confined. This sequence is supported experimentally for the spectrum and numerically for transport and localization.

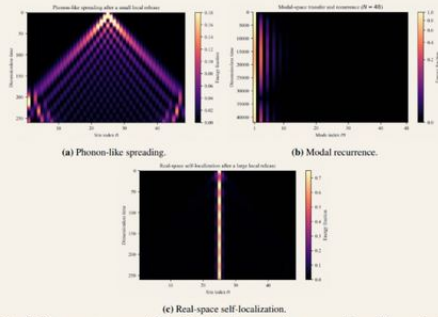


Fig. 4 · Three outcomes (N = 48): (a) phonon-like spreading, (b) modal recurrence, (c) real-space self-localization.

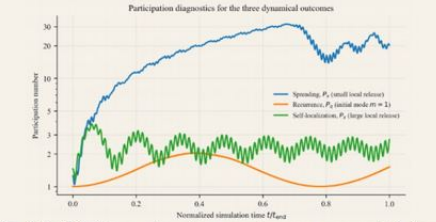


Fig. 5 · Participation numbers: P_x grows when energy spreads, P_q rises and returns during recurrence, P_x stays at 2–3 sites when localized.

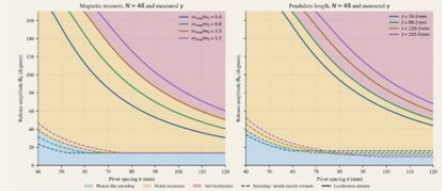


Fig. 6 · Estimated boundaries in the (s, θ_0) plane: blue spreading, yellow recurrence, red self-localization.

100+ trials on my own 3D-printed rig

2k+ audience for my research talk

30+ forums and inter-school exchanges led

10+ workshops run

200+ hours coaching

20+ students coached

Electrical *Damping*

Problem. *A magnet suspended by a spring will display simple harmonic motion when displaced. If the magnet oscillates within a coil connected to a resistor, its motion will be damped. Investigate the factors that affect the damping.*

1 · THE PHYSICS

A magnet hanging on a spring oscillates inside a coil wired to a resistor. As it moves, the changing flux induces an emf; the induced current feels an Ampère force that pushes back on the magnet, and its energy leaves as Joule heat in the resistor. Damping is therefore a chain: mechanical energy → electrical energy → heat. In a preliminary run (R = 10 Ω, N = 400 turns, μ = 27.02 A·m²) the amplitude decayed exponentially, and damping was slightly stronger when the magnet was near the coil's edge.

2 · MODEL

I modelled the bar magnet as an equivalent current sheet (measured magnetisation M = (1.12 ± 0.04) × 10^6 A/m) and integrated Biot–Savart over the coil to get the flux ψ(x, N, B) and the mutual-inductance gradient G = dψ/dx, so ε = G ẋ. Kirchhoff and Newton then couple the circuit to the motion:

$$G\,\dot{x} = L\,\dot{I} + I\,R$$

$$m\,\ddot{x} = -k(x - x_0) - mg + I\,G - 2m\gamma_{mech}\dot{x}$$

ignoring inductance: $F_{em} = -(G^2/R)\,\dot{x},\ \gamma_{em} = G^2/R$

Solved with RK4, the motion is underdamped; an FFT of the current shows several superposed frequencies, because the magnet's field has strong nonlinear terms.

3 · PREDICTIONS

Turns: damping rises quickly up to a peak at 2,700 ± 10 turns, then slowly falls to 6,000 turns as the coil's own inductive reactance chokes the current. Resistance: damping falls roughly as 1/R. Magnetic moment: since ε ∝ m and F ∝ I·m, damping first grows quadratically, then levels off as the heavier magnet gains inertia.

4 · EXPERIMENT



Fig. 1 · My rig: spring and N52 magnet, coil, resistance box, galvanometer, displacement sensor.

Spring 16 N/m; total mass 0.05 kg; N52 cylinder magnet 8 mm × 12 mm; coil 2,666.67 turns/m. I varied turns (200, 800, 1,600), magnets (2–16) and resistance (2–10 Ω), tracked the motion on video with Tracker and logged current. With the circuit open, mechanical damping was γ_{mech} = 0.0162 ± 0.0005; at Re ≈ 183 air drag is linear, so it could be subtracted cleanly.

5 · RESULTS

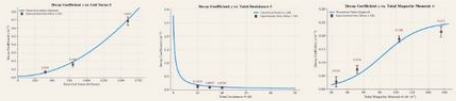


Fig. 2 · γ against turns, resistance and magnetic moment: model (line) and my data (points).

All three trends match the model. Errors: a 3.6% magnetisation error doubles to 7.2% in damping (F ∝ M²); inductance carries 10%; large amplitudes push the spring past linearity, and a slight swing or spin of the magnet changes dψ/dt.

2,700
turns: predicted damping peak

1/R
damping vs resistance, confirmed

7.2%
largest error, traced to M²

Three problems, *one team*

Besides the magnetic cradle on the previous page, I reported these two problems myself. As captain I led our team's discussions: we took each problem apart into physics, model and experiment, argued over every assumption, and rehearsed the opposition and review rounds together before the tournament.



Leading a team discussion before the fights.



Reporting Electrical Damping.



The whole team at CYPT 2026; I am the one with the red hair tie.

Levitation *Control*

Problem. *When arranged in a specific configuration, small graphite sheets can levitate on neodymium magnets. By shining light onto the surface of the graphite sheet, it is possible to control its movement. Explain and investigate the phenomenon.*

1 · THE PHYSICS

Pyrolytic graphite is diamagnetic, so it floats above the potential minimum of a checkerboard of NdFeB magnets. A laser spot heats one edge; graphite's susceptibility depends on temperature, so the force on the sheet becomes uneven and it tilts and slides. Depending on laser power and speed, it stays levitated, relaxes back, or follows the light continuously.

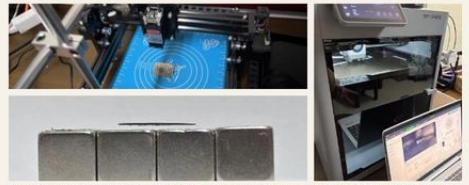


Fig. 3 · Equipment I bought and set up: the laser on a 2-D gantry, my 3D printer making holders, and graphite floating on my magnets.

2 · FIELD AND FORCE

For an m × n array of alternating cubic magnets I built the field from surface currents (J = 3.651 × 10^5 A/m²) by Biot–Savart and superposition. With anisotropic χ (χ∥, χ⊥) and χ ≪ 1, the energy density is u = −χB²/2μ₀. Levitation balances ρg = (χ⊥/μ₀) B_z ∂B_z/∂z; the periodic array term cancels over a cycle, leaving the light-driven term:

$$F_r = (h / 2\mu_0) \cdot (d\chi_{\parallel}/dT) \cdot (dT/dr) \cdot \iint B_{xy}^2\, dA$$

From the Landau free energy, χ depends on temperature; I used the measured $\chi_\perp = (-743 + T) \times 10^{-6}$ and $\chi_\parallel = (-135 + 0.25T) \times 10^{-6}$.

3 · TEMPERATURE FIELD

A 3-D transient heat equation with a Gaussian spot gives the surface temperature. At 300 mW the gradient is constant from 0.05 s on, with ΔT = 1.36 K across the sheet.

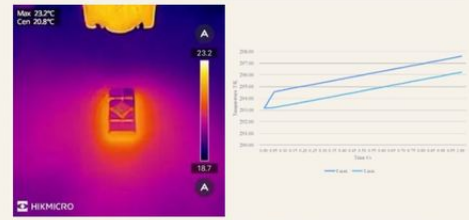


Fig. 4 · My thermal-camera image of the heated sheet; model: hottest and coolest points under 300 mW.

4 · MOTION

Newton's law m f̈ = F_r − F_k − ζḟ combines light drive, a restoring force from the array's potential (periodic with the magnet edge l) and thin-film air drag. If $F_r \le F_{k,max}$ + drag the sheet relaxes within one period l; above it, it moves continuously.

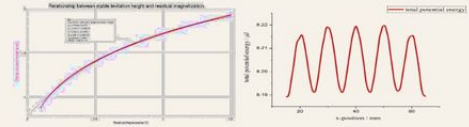


Fig. 5 · Model: levitation height against remanence; the potential along the array repeats with the magnet edge l.

5 · EXPERIMENT, HONESTLY

Magnets of 10 and 5 mm at 0.1–0.4 T, graphite 0.3 mm and 50 μm, a 0–1.6 W laser on a 2-D gantry and a thermal camera. The sheet levitated and heated as predicted, but my set-up did not yet reproduce light-driven travel. My analysis: N52 magnets hold the sheet too firmly for a 1.6 W beam. Next: weaker magnets, thinner graphite, more power.

1.36 K
light-made gradient across the sheet

4
coupled models: field, χ(T), heat, motion

1.6 W
laser limit I must beat next

SmartHearing: *Scene-Adaptive Hearing Assistance* via LALM-Guided Language-Queried Sound Enhancement

Research question. *Which sound matters most right now, and how can a hearing aid enhance it in a scene it has never heard before?*

XIANGYU KONG & AMY (WANYING) HU, EQUAL CONTRIBUTION · ICASSP 2027 SUBMISSION

ABSTRACT Conventional hearing aids prioritize speech, yet which sound matters most (a siren, an alarm, a scream) depends on the acoustic scene. Speech-centric models suppress non-speech warnings, and scene classifiers rely on predefined labels that do not transfer to unseen environments. We propose a scene-adaptive hearing assistance framework that couples a large audio language model (LALM) for open-vocabulary scene understanding with language-queried sound separation. A lightweight Query-space Adapter bridges the representation shift between LALM-generated descriptions and the separator's training queries, and an adaptive window selects 4–10 s of context per sound category. On 600 mixtures, the Adapter improves SNRi and SI-SDRi by 0.97 dB and 1.14 dB over the base separator; on 145 unseen ESC-50 mixtures, the gains persist (+1.32 dB, +1.19 dB). In a 101-participant listening test, the system was rated most salient in 50.5% of trials versus 28.1% and 21.5% for baselines.

MY ROLE

AI hearing-aid product design lead and equal-contribution author; I hold a patent for the hearing aid, and my startup develops it as a product (p. 20).

WHY IT MATTERS TO ME

My grandfather's hearing challenges taught me to notice quiet details. This project is about the loud ones that must never be missed.

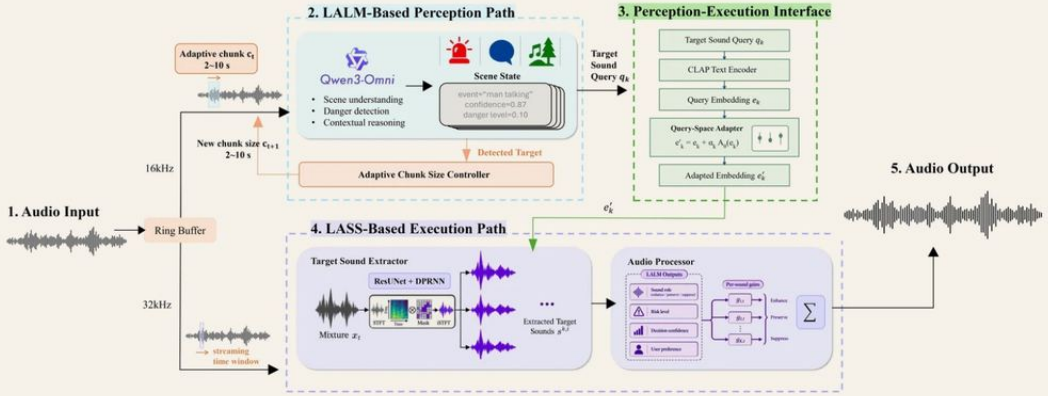


Fig. 1 · Overall architecture: an LALM-based perception path and a LASS-based execution path, joined by the Query-space Adapter.



Fig. 2 · The target depends on the scene: a siren, speech in a restaurant, a crying infant, a horn.



Mean overall comfort did not differ significantly ($p = 0.074$): salience is not yet listening comfort.

Fig. 3 · Listening test, 101 participants (303 scene-level choices): share of trials where the target was easiest to notice.

1 · MOTIVATION

Hearing-impaired listeners regularly meet restaurants, shopping malls and multi-speaker meetings, where background noise, mixed sounds and overlapping speech coexist. Conventional hearing aids primarily amplify human voices, so other sounds that carry critical information, such as a car horn on the street, may be attenuated, creating fatal risk.

2 · METHOD

The perception path uses Qwen3.5-Omni to decompose each 16 kHz window into sound objects o_k = (sound, role, danger, confidence, relevance), with six roles from *urgent warning* to *unwanted noise*. A frozen CLAP encoder maps each description to a 512-d embedding, and the Query-space Adapter corrects it: $e'_k = e_k + \alpha_k A_\theta(e_k)$. LASS (ResUNet + DPRNN) extracts each source at 32 kHz, and an Audio Processor sets gains by risk and confidence: $y_t = g_b b_t + \Sigma\, g_k s_k$. The two paths run asynchronously, so inference never blocks audio.

3 · EXPERIMENTS

Adapter V1 is trained on FSD50K; V2 adds ESC-50, UrbanSound8K, NIGENS and TUT Rare Sound Events 2017. Each of seven target classes contributes 500 training and 100 validation mixtures, with ten interferers and SNRs from −10 to 10 dB.

4 · RESULTS

Speech denoisers collapse on non-speech targets (−18.4 dB SI-SDRi for GTCRN) and a fixed query collapses entirely. Prompt engineering alone (10.21 dB) matches base LASS, so the Adapter's gain cannot be replicated by query design. With no ground-truth label, SmartHearing recovers most of the oracle gap. The Adapter raises within-class similarity from 0.785 to 0.860 and cuts the distance to LASS training queries from 0.181 to 0.119. For a siren in wind, residual energy falls 7.05 dB and SI-SDRi rises from 0.16 to 7.21 dB.

An ablation isolates the Adapter: V1 already improves on base LASS, and V2, trained on a larger mixture set, significantly outperforms V1 in most metrics (ΔPESQ +0.61 [0.25, 0.96]). Between-class similarity also rises, from 0.337 to 0.485, which reflects shared convergence toward the region LASS expects rather than a loss of discrimination.

5 · CONCLUSION

A proper Adapter between perception and execution improves separation of the sounds that matter. Next: cross-checking LALM outputs and evaluation with hearing-impaired listeners.

For a hearing-aid user this means a siren, an alarm or a baby's cry is no longer treated as noise to be removed, even in a scene the system has never been trained on.

Method	Scene prior	SNRi (dB)	SI-SDRi (dB)	PESQ
Oracle query + LASS (upper bound)	✓	12.45	11.29	2.64
Semantic Hearing TSE	✓	2.63	−1.43	1.56
Wiener filter	–	−0.64	−0.75	1.48
GTCRN speech denoiser	–	−2.42	−18.38	1.30
Fixed query + LASS	–	−2.55	−22.81	1.45
Prompt engineering + LASS	–	10.21	9.01	2.47
Qwen3.5-Omni + base LASS	inferred	10.24	8.99	2.52
Gemini 3.5 Flash + LASS	inferred	10.37	9.11	2.55
SmartHearing	**inferred**	**11.24**	**10.32**	**2.60**

Table 1 · Scene-adaptive performance, mean across scenes.

6 · LATENCY AND WINDOWS

Inference is dominated by the LALM API (median 1.24 s) against only 0.067 s of local separation. A window sweep over ten lengths, six safety sounds and ten interference types (6,000 units) showed that no single window is best: a bark needs about 4 s, a siren or a crying baby 10 s.

1
patent for the AI hearing aid

8.7k+
audio samples across 5+ datasets

300+
listener trials

IEEE
1st author paper, ICASSP

→ 17
taken to market by my startup

Flowed Metal: *Micro-particle Impacts on Metallic Glasses* under High Temperature

Research question. *When a metallic glass is hit at hundreds of metres per second, does heat make it harder, like a metal, or softer, like a glass?*

AMY (WANYING) HU, FIRST AUTHOR · SUPERVISOR PROF. MINQIANG JIANG · SPACECRAFT SOLID MECHANICS RESEARCH INTERN

ABSTRACT Bulk metallic glasses (BMGs) represent a unique class of materials that combine the disordered atomic structure of glass with the properties of metals. Their response to dynamic loading, such as high-velocity impact, is not only crucial for potential structural applications but also fundamental to understanding non-crystalline physics. Crystalline metals show a counterintuitive phonon-drag strengthening effect under high strain rates at high temperatures, which contrasts with glasses' conventional thermal softening behaviour. Therefore, whether BMGs exhibit a similar phenomenon remains an open question. In this work, the impact response of a $Pd_{43}Ni_{10}Cu_{27}P_{20}$ BMG at 25 °C, 100 °C and 200 °C was investigated via laser-induced particle impact tests and finite element method simulations. In-situ observation revealed a significant enhancement in energy dissipation of BMGs at 200 °C. Post-impact characterization identified a transition in deformation mechanism: solid-like craters with minimal pile-up at 25 °C gave way to fluid-like, irregular pile-up morphologies at 200 °C, suggesting adiabatic heating-induced softening. A calculated ~10.4% reduction in dynamic yield strength at high temperature provided quantitative confirmation of this thermal softening. Furthermore, simulations incorporating a physical constitutive model successfully replicated the experimental results and uncovered the underlying mechanism: high temperature promotes the activation of shear transformations, leading to greater plastic flow and prolonged impact interaction time. This study conclusively demonstrates that temperature is a critical, governing factor for the dynamic performance of BMGs, challenging simplistic material selection and providing vital insights for their application in extreme environments. *Key words: micro-particle impact, metallic glass, temperature, plastic flow, energy dissipation.*

1 · BACKGROUND

Glass is a solid at room temperature but behaves like a super-viscous liquid when hot, because its atoms are arranged with "short-range order and long-range disorder". Metals are crystalline, with long-range periodic order that gives them strength and ductility. Metallic glass combines the two: molten metal is cooled so fast that its atoms have no time to crystallise, freezing the disordered liquid structure into a solid.

Under impact, ordinary glass shatters while metal deforms plastically. Recent work showed that at strain rates above 10^4 s^{-1} pure metals (Cu, Au, Ti) become *stronger* as temperature rises, through phonon drag. Would a metallic glass harden like a metal, or soften like a glass?

2 · SAMPLE

$Pd_{43}Ni_{10}Cu_{27}P_{20}$ was chosen for its low critical cooling rate and high amorphous stability. Samples (10 × 5 × 3 mm^3) were wire-cut with a diamond saw, ground, polished and mounted on a heated glass plate. XRD showed only one broad hump near 40° and no sharp peaks, confirming the glassy state; DSC gave T_g = 570 K and crystallisation at 632 K, so 25–200 °C keeps the sample amorphous.

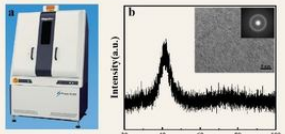


Fig. 1 · a XRD set-up; b one broad amorphous hump, TEM inset.

3 · LIPIT PLATFORM

A 7 ns, 532 nm pulsed laser is focused through a convex lens onto a launch pad: 4 mm K9 glass, a 100 nm gold ablation layer and an 80 µm polymer elastic layer carrying 30 µm SiO_2 particles. The expanding plasma flings a particle at 1–1000 m/s; a camera at up to 5,000,000 fps records it. A REX-C900 PID controller holds the target from 25 to 400 °C within ±1 K. Craters were then imaged by SEM and their depth profiled by laser confocal microscopy.

4 · ENERGY DISSIPATION

From impact speed v_i and rebound speed v_r, COR = v_r/v_i. At 200 °C the COR is consistently lower: the hot glass dissipates more energy, the cold glass resists penetration better. Between 80 and 250 m/s both follow COR ∝ $v_i^{-0.5}$ (finite plasticity: 8.6 $v_i^{-0.5}$ at 25 °C, 7.7 $v_i^{-0.5}$ at 200 °C); above 250 m/s they fall more steeply, 140 v_i^{-1} against 372 $v_i^{-1.2}$.

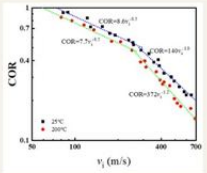


Fig. 2 · COR against impact velocity, 25 °C (black) and 200 °C (red).

5 · DYNAMIC YIELD STRENGTH

In the finite-plastic regime, Wu's relation COR = $\alpha[(v_i/v_y)/(E^*/Y_d)]^{-1/2}$ with α = 0.78 and Johnson's yield velocity v_y can be solved for Y_d. The result: 2089 ± 110 MPa at 25 °C, about 1901 MPa at 100 °C and 1825 ± 65 MPa at 200 °C, a fall of **10.39%**. The glass softens: it trades impact resistance for energy dissipation.

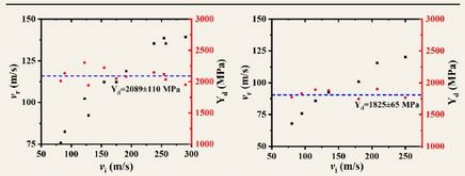


Fig. 3 · Y_d at 25 °C (left) and 200 °C (right).

6 · SIMULATION

To see the mechanism, I built a quarter-symmetric ABAQUS model (about 1.35 million C3D8I elements, 0.25 µm at the contact) with a coupled shear-transformation-zone / free-volume constitutive model. Its crater profile matched the confocal measurement at 300 m/s. At 400 m/s the deepest indentation came at 14.4 ns (25 °C) against 15.2 ns (200 °C), and separation at 23.6 against 25.6 ns: a longer contact, lower stress, greater plastic strain, and many more positive shear transformations at 200 °C.

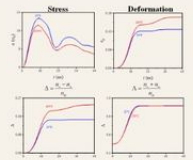


Fig. 4 · Field-averaged stress, plastic strain and STZ densities; red 200 °C, blue 25 °C.

7 · LIMITATIONS

Yield stress is inferred, not measured, so small velocity errors grow through v^2. Thermal gradients mean the true impact temperature is unknown, and adiabatic heating could not be measured. Above T_g the surface oxidised badly (we tried a vacuum, but the chamber needed was too large), so the range stopped at 200 °C.

8 · CONCLUSION

Temperature governs how a metallic glass meets an impact: from solid-like to fluid-like, with 10.4% lower dynamic yield strength and more energy absorbed. Unlike crystalline metals, it does not harden when hot.

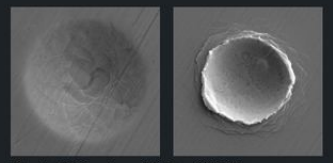


Fig. 5 · SEM craters, 25 °C vs 200 °C.

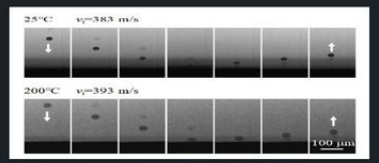


Fig. 6 · High-speed frames, 25 °C and 200 °C.

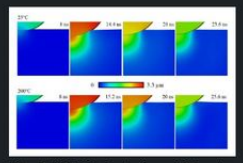


Fig. 7 · FEM indentation, 400 m/s.

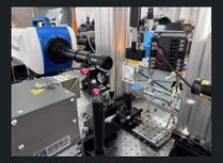


Fig. 8 · The LIPIT rig I ran.

MY ROLE

First author. I prepared and characterised every sample, ran the LIPIT impacts at all three temperatures, computed COR and yield strength, and built the ABAQUS simulation with a PhD mentor in Prof. Minqiang Jiang's group.

WHY I ASKED

"In middle school I learned that glass has no melting point. When hit, would a metallic glass shatter like glass, or flow like metal?"

100+ impact tests on metallic glass

5 Mfps high-speed camera

3 temperatures: 25, 100, 200 °C

−10.39% dynamic yield strength

1.35 M elements in my FEM model

ISEF Beijing qualifier · first author

Curawave: A *Wave-Bionic Soft Robotic Mattress* for Post-Operative Patient Transfer

Design question. *Can a mattress slide a patient sideways onto another bed the way a wave moves a boat, safely and without anyone lifting?*

AMY HU, CTO · WITH EMILY FENG (CEO) AND GALATEA SHANG (CFO)

AS CTO AND FOUNDER

I led the technology: mould design, silicone casting, valves and Arduino control. My startup, Beijing Lingtong Future Technology, raised over USD 100k to develop assistive products (p. 20).

WHY I BUILT IT

The idea began with my own bicycle accident. With an injured leg I struggled to move from place to place, and every attempt was shocked by pain. Moving patients after surgery is the same problem at hospital scale: in 2024 China had about 70 million people bedridden by injury and 96.4 million surgical operations.



Fig. 1 · From our pitch deck: the brand and channels, the product, the market, five-year finances, strategy and the marketing plan we built as a team.

SUMMARY Among 1,200 post-operative patients in a study we cited, 15% experienced complications related to being transferred, and different moving devices and transition times influence secondary injury. Curawave is a mattress that imitates the movement of ocean waves to transfer patients safely and comfortably without human labour. It is built from 18 × 20 small soft units, each with a hollow silicone inflation layer, a non-stretchable cotton confining layer and a silicone base. Because the confining layer stretches less than the inflation layer, each unit tilts sideways, left or right, when inflated; adjacent units are grouped and inflated alternately, so a sideways wave slides the patient horizontally off one bed and onto the next, with no one lifting. We modelled the moulds in Autodesk Fusion, 3D-printed them, cast the silicone units, and drove an air pump through solenoid valves controlled by an Arduino, producing a working 1 × 4 prototype.

MARKET & PLAN

China's medical-device market is about 120 billion yuan. We start with first-tier hospitals: direct sales, training videos and yearly maintenance. Forecast turnover grows from ¥2.3M to ¥21.6M in five years, with 1% to charity.

NEXT STEPS

AI that reads the forces on a patient and adjusts each unit's inflation; a patent on how the units are assembled to mimic waves; and markets beyond mainland China.

① MOULD MODELLING

Autodesk Fusion: two bases, top and bottom, and an inner core.

② ONE UNIT, TESTED

Moulds 3D-printed; silicones of different thickness poured and compared.

③ VALVES + CODE

Solenoid valves open and close the air pump; an Arduino program times them.

④ 1 × 4 MODEL

The main pump's air is split four ways, each group switched by its own valve.

⑤ FURTHER TESTING

Units tested in sequence to tune timing and inflation.

INNOVATIONS Transfers patients sideways from bed to bed with no lifting · massages patients who cannot move, such as those in a vegetative state · waterproof, bacteria-resistant and easy to clean · lower price than existing transfer devices.

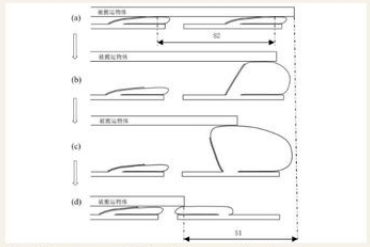


Fig. 2 · How one unit tilts sideways and pushes.



Fig. 3 · Cast silicone units.

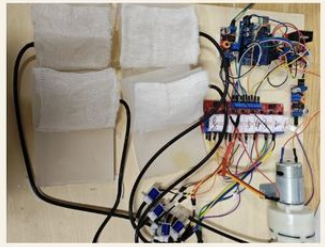


Fig. 4 · Pump, valves and Arduino.

$100k+ raised for my assistive-tech startup

3+ wave-mattress prototypes

10+ CAD and circuit designs

20+ parts 3D-printed or cast

100+ people pitched to

2 products: hearing aids & mattress

Beijing Lingtong *Future Technology*

Mission. *A human-centred assistive technology company: turning research on hearing and patient care into products people can buy, use and trust.*

FOUNDED BY AMY HU · A TEAM OF ABOUT FIVE · USD 100K+ RAISED

MOTIVATIONS A paper proves an idea works; only a product reaches the person who needs it. Both of my engineering projects began with someone I know: my grandfather, who struggled to hear, and my own weeks on an injured leg. Research at Peking University and the Conrad Challenge gave us working prototypes, so I founded Beijing Lingtong Future Technology to carry them further. The company keeps one small R&D team behind both products, so what we learn about sensors, control and testing on one is reused on the other. Each co-founder owns one side of the business, and every spending decision is checked against our five-year model. As founder and CTO, I lead the technology for both products and pitch them to investors and hospitals.

Lingtong

HOW THE COMPANY IS ORGANISED

LINGTONG FUTURE TECHNOLOGY
One shared R&D team behind two product lines.

FOUNDER & CTO · ME
Technology, prototypes and patents for both products.

CEO
Strategy, partners and the go-to-market plan.

CFO
Budget, financial model and fundraising.

ALGORITHMS & AI
Sound understanding, priority decisions, separation.

EMBEDDED & HARDWARE
Real-time audio, low power; valves and control.

PRODUCT & DESIGN
Wearable form, CAD, 3D printing, casting.

REGULATORY & CLINICAL
Device classification, safety, pilots.

BUSINESS & SUPPLY
Partners, ODM makers, channels.





TRADEMARKS

Filed on 1 July 2026 with China's National Intellectual Property Administration (CNIPA). Application No. 92712127 (Class 42, scientific and technological services) was accepted on 20 July 2026. Application No. 92712148 passed preliminary examination and was published in Trademark Gazette No. 2001 on 27 September 2026.

Left: acceptance notice. Right: preliminary approval notice.

TWO PRODUCT LINES



SmartHearing
Core product: an AI hearing system that decides which sound you need to hear. Patent filed; p. 17.



Curawave
Second growth curve: a wave-bionic transfer mattress, three prototype generations; p. 19.

Focus: 70–80% of resources go to SmartHearing for the first 18 months.

BUSINESS MODEL · THREE REVENUE STREAMS

1 **Own-brand devices**
SmartHearing as device + app + long-term service; basic hearing never put behind a subscription.

2 **Institutional solutions**
Hearing centres, care homes and hospitals: equipment, training and maintenance. The mattress is B2B only.

3 **Technology licensing**
An Auditory Decision Engine SDK for earbuds, hearing aids and smart glasses.

First users: active adults 50+ with mild-to-moderate hearing loss, often bought by their adult children.

THREE-YEAR PLAN

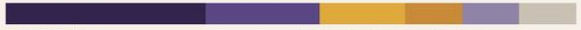
0–6 mo	Research prototype → wearable SmartHearing; final mattress prototype; device-classification advice.
6–12 mo	Pilots with 100–300 users and 3–5 hearing, care or rehab partners; first paid B2B pilots.
12–24 mo	Design freeze, registration path, supply chain and first paid sales.
24–36 mo	National channels; licensing of the Auditory Decision Engine begins.

Planning case: revenue, ¥ million (a target to plan against, not a promise)



FUNDING & PARTNERS

First external round planned at **¥5–8 M** for about 18 months, spent on engineering, not advertising:



- 35% R&D
- 20% Prototypes & tooling
- 15% User & institution pilots
- 10% Regulatory, testing, IP
- 10% Key hires
- 10% Reserve

Pilot partners: hearing centres, fitting clinics, ENT and rehab institutions, elderly-care communities

Manufacturing: ODM/OEM makers with hearing-aid, earbud and medical-electronics experience; rehab-equipment makers for the mattress

Advisers: audiologists, clinical advisers and older users inside the design loop

$100k+
raised so far

~5
people on the team

¥5–8 M
first external round planned

1.5 bn
people with hearing loss (WHO)

323 M
Chinese aged 60+, end of 2025

<5%
hearing-aid penetration in China

1 · GEOMETRIC OPTICS

With the conjugate method the focal length of a convex lens came out at 100.0 mm against a nominal 100 mm. The method only needs the screen distance D and the lens shift d, $f = (D^2 - d^2)/4D$, which cancels the uncertain position of the lens's optical centre. Three screen distances (550, 600 and 650 mm) gave consistent results.

A calibrated eyepiece micrometer gave 8.40 µm per division and a fibre diameter of 71 µm; the scales had to be recalibrated for every objective. At minimum deviation, a glass prism gave $n \approx 1.517$ and showed dispersion: violet bends most, red least. Finding minimum deviation means watching the spectral line reverse direction, so I read each position several times.

2 · INTERFERENCE

In Newton's rings the air film between a lens and a flat plate produces dark rings with $D_m^2 = 4mR\lambda$. Using two widely separated orders removes the uncertain contact point, and gave $R \approx 0.85$ m. I always turned the micrometer drum in one direction to avoid backlash.

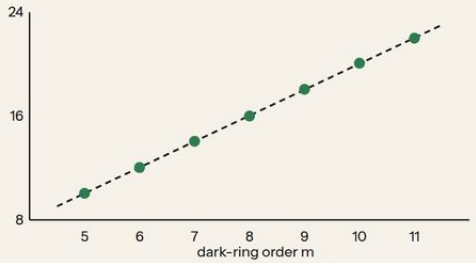


Fig. 1 · My Newton's-rings data: D^2 grows linearly with ring order m, as predicted.

An air wedge gave a sheet thickness of about 74 µm by measuring 20 fringe spacings at once; pressing the glass made the fringes visibly move. Young's double slit gave a He-Ne wavelength of about 633 nm from ten fringe spacings, and the Michelson interferometer showed rings appearing from and collapsing into the centre as one mirror moved.

3 · DIFFRACTION

Single-slit Fraunhofer diffraction gave a slit width of 0.20 mm: the central maximum is twice as wide as the others, and the whole pattern widens as the slit narrows. A transmission grating, read on both sides of zero order to cancel offset error, gave the mercury green line at about 546 nm.

4 · POLARISATION

Rotating an analyser in 15° steps reproduced Malus's law, $I = I_0\cos^2\theta$, almost exactly; the small intensity left at 90° comes from the finite extinction ratio of real polarisers and stray light.

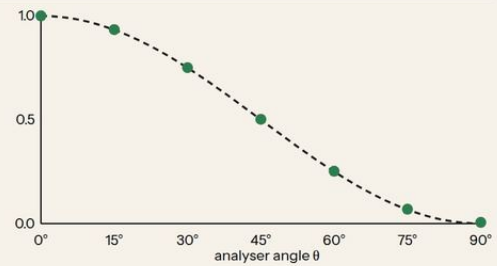


Fig. 2 · My Malus's-law data (dots) against $\cos^2\theta$ (dashed).

5 · ALL TWELVE REPORTS

Experiment	Measured	Result
Thin-lens focal length	conjugate method	f ≈ 100.0 mm
Microscope calibration	fibre diameter	≈ 71 µm
Prism, minimum deviation	refractive index	n ≈ 1.517
Newton's rings	lens radius	R ≈ 0.85 m
Air wedge	sheet thickness	≈ 74 µm
Young's double slit	He-Ne wavelength	≈ 633 nm
Single-slit diffraction	slit width	≈ 0.20 mm
Diffraction grating	Hg green line	≈ 546 nm
Malus's law	I/I_0 at 45°	0.502
Michelson interferometer	λ from counted fringes	—
Abbe refractometer	water · solution	1.3330 · 1.3474
Laser holography	3-D image	reconstructed

Table 1 · Results of my twelve lab reports.

6 · REFRACTION & HOLOGRAPHY

An Abbe refractometer gave 1.3330 for distilled water and 1.3474 for a sugar solution, recording temperature because index changes with it. In static laser holography, the developed plate looked like noise, but lit by the He-Ne beam it rebuilt a 3-D image with real parallax: proof that the plate stored phase as well as amplitude.

7 · WHAT THE ERRORS TAUGHT ME

Every report ended with an error analysis, and the same lessons kept returning: judge the sharpest image or the darkest fringe several times; measure many fringes at once rather than one; read both sides of a symmetric pattern to cancel offsets; turn screws in one direction only to avoid backlash; and record temperature, because it changes refractive index. Those habits are what made the results close to the reference values.

8 · INTO THE RESEARCH LAB

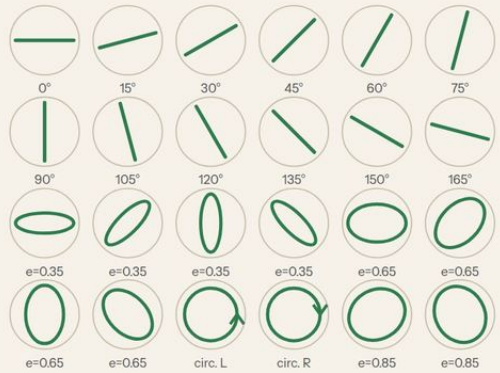


Fig. 3 · Polarisation states of the kind I tested: linear every 15°, elliptical and circular.

For holography I recorded with the 632.8 nm He-Ne beam at about 5 mW, an object distance of about 25 cm, a reference angle of about 35°, an object-to-reference intensity ratio of about 1 : 1.5 and a 5 s exposure; even a fraction of a wavelength of vibration during exposure would have erased the fringes.

12
optics experiments, each a full lab report

20+
polarisation states tested

Raman
and laser systems, graduate level

0.4%
Malus's law deviation at 45°

Nobel
laureate conversation

PEKING UNIVERSITY, INSTITUTE OF PHYSICS

Polarisation Control and *Experimental Optics*

Aim. *Learn to measure light precisely, then control its colour, intensity and polarisation in a research lab.*

AMY HU · NONLINEAR OPTICS RESEARCH INTERN · 12 WRITTEN LAB REPORTS

SUMMARY I worked through twelve classic optics experiments, from the focal length of a thin lens to static laser holography, and wrote each up as a full lab report with aims, principle, procedure, data and error analysis. The measurements agree closely with reference values: a He-Ne wavelength of about 633 nm against 632.8 nm, the mercury green line at about 546 nm against 546.1 nm, and a normalised intensity of 0.502 at 45° against the $\cos^2\theta$ prediction of 0.500. In the research lab I then operated graduate-level Raman and laser systems, tested more than twenty polarisation states of the beam, and spoke with a Nobel laureate about where experimental physics is heading.

INTO RESEARCH

These skills carried straight into the research lab. There I operated Raman and laser systems normally used by graduate students, and tested more than twenty polarisation states of the beam, from linear at every 15° to elliptical and circular. A conversation with a Nobel laureate showed me where experimental physics is heading next.

IN THE LAB



My sample, the clean room, an optical table I aligned, and the laser and spectrometer bench. I built every set-up myself: aligning the optical axis, calibrating scales and repeating each measurement.

Mathematical Modelling *and Science Publishing*

Approach. *Turn a messy real problem into variables, assumptions and a model we can test, then explain it clearly.*

AMY HU · TEAM CAPTAIN AND MAIN MODELLER · AUTHOR OF ALL FIVE PAPERS · 3 FINALIST AWARDS

SUMMARY I wrote all five papers below, as team captain and main modeller, turning open problems into tested models: the optimal scale and cost of water-saving basins for the Panama Canal; the shape of a spinning jump rope for a humanoid robot; optimal sweep routes for responders in a burning building; whether Friday the 13th is really unlucky; and patrol routes and schedules for Etosha National Park. Three of the papers received Finalist awards. As captain I split each problem into sub-models, built and coded the core model myself, and wrote and edited the final paper within the contest deadline.



A SEPARATE ROLE: JOURNAL EDITOR

I am also an editor of our school's mathematics journal. Our Vol. 2, *Applied Mathematics*, was an IMMC issue: it explained contest models, including ours, for every reader, alongside interviews with alumni and researchers.

5	Captain	3×	Vol. 2	200+
papers, all written by me	and main modeller	Finalist awards	journal issue on IMMC	journal readers

IMMC 2024 · PROBLEM A

Saving the Panama Canal's Water

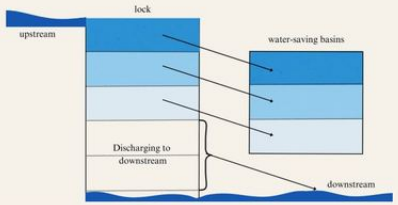


Fig. 1 · Water exchange between the lock and its water-saving basins.

Without water-saving basins, lock chambers continuously transfer water from upstream to downstream, leading to upstream scarcity, drought and navigation risk.

From the water-exchange diagram between basins and chambers we derived how saved water depends on the number of basins and their size. This gave the optimal basin scale, so no saving potential is wasted.

We then combined energy, maintenance, water, labour and material costs into one overall cost model and, with local data, found the economically optimal design and its annual cost.

OPTIMISATION · COST MODEL

IMMC 2025 · PROBLEM D

Teaching a Robot to Jump Rope

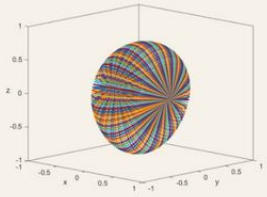


Fig. 2 · The rope's 3-D trajectory through one full turn.

A company asked us to develop a rope-jumping function for its humanoid robot, FlexRobot. We first analysed the robot's motion on the ground and in the air.

We treated the rope as a catenary, and noticed that the shape of a rotating catenary can be described by the Jacobi elliptic function sn. We built each of its parameters as a function of time and drew the rope at any moment of a jump in 2-D and 3-D in MATLAB.

Finally we computed the ranges of initial jump velocity, jump time and rope length that make a successful jump possible.

CATENARY · JACOBI SN · MATLAB

HIMCM 2025 · PROBLEM A · FINALIST

Sweeping a Burning Building

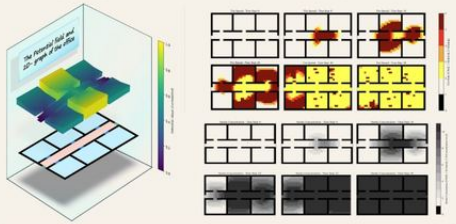


Fig. 3 · Potential field of an office floor (left); my simulated fire (top) and smoke (bottom) spreading over time.

To plan the best route for responders sweeping a building in an emergency, we minimised total room-clearing time while prioritising occupants' safety.

A cellular automaton models fire spread, smoke spread, occupant movement and room importance as potential fields; occupants' speed varies with age and crowd density. Mixed-integer linear programming with CP-SAT then finds the route and number of responders that scan every cell in the shortest time.

We applied it to an office, a restaurant and a school lab, extended it to gas leaks, and tested it with the Fourier Amplitude Sensitivity Test.

CELLULAR AUTOMATON · MILP

IMMC 2026 · PROBLEM D

Is Friday the 13th Unlucky?

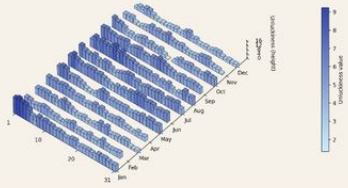


Fig. 4 · Predicted "unluckiness" of every date in 2025.

A severity model combines casualties and economic loss for 2000–2025 disasters. An expectation model, an analogy of the Boltzmann distribution with "temperature" replaced by people's level of expectation, gives a Disaster Severity Expectation distribution; information theory turns the gap into surprisal.

The maximum unluckiness was 18.21 and the median 4.42; Friday the 13th averaged 5.56, with no peak, so the superstition is disproved.

A graph model with Greedy Clique Expansion predicted every date in 2025: all values lie within 7–8.

BOLTZMANN · ENTROPY · GRAPH CLIQUES

IMMC 2026

Guarding Etosha National Park

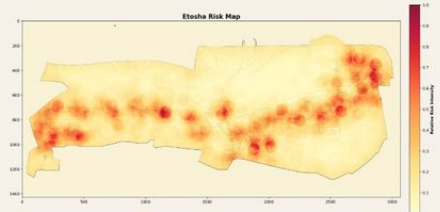


Fig. 5 · Risk map of Etosha National Park.

Etosha sets an immense area against a small ranger team. Our framework runs from risk mapping through topological decomposition and geometric routing to scheduling and staffing.

A risk model merges human threats and natural volatility into one metric. Topological decomposition splits the park into traversable patrol units. Inside each, an Active Contour (Snake) algorithm draws a smooth closed loop, and outward-bulging "petals" reach remaining high-risk clusters.

The Nyquist–Shannon theorem sets a baseline patrol frequency, and the Kalman-filter steady state gives a closed-form safety-critical revisit interval.

RISK MAP · SNAKE ROUTING · KALMAN

SOUND





Duet on stage



Ensemble, outdoors



Singing with friends

Guzheng · voice

I play the guzheng, the 21-string Chinese zither, as a soloist, in duets and with an ensemble, and I sing on the school stage. I performed at our school gala in the spring of Grade 10 and at the school's public performance in the autumn of Grade 11.

MOTION





Competition bout



Teaching young students at Shangdi Youth Taekwondo Academy

Taekwondo · black belt · Beijing gold

Eight years of taekwondo and a Beijing gold medal. Since Grade 5 I have also been an assistant instructor at Shangdi Youth Taekwondo Academy, coaching 50+ younger students in forms, kicks, sparring and discipline.